

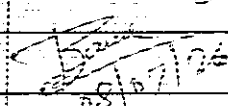
Name : Aarush Hemant Sawant

Class : T.Y.B.Sc. (T.T.)

Roll No. :

Subject : Internet of Things (IoT) Assignment

- Index

Sl. No	Date	Topic	Sign
1	08/07/2026	The Internet of Things : An Overview & Design Principles	 08/07/26
2		Prototyping Concepts & Embedded Device	
3		Prototyping Physical & Online Components	
4		Embedded Code Techniques & Business Models	
5		Manufacturing Processes & Ethical Consideration	

[illegible]

Sr.No.	Date	Topic	Sign
1	8/7/26	The IoT: Overview & Design Principles	<i>Bawa</i>

Q.1 ~~IoT~~ What is IoT, give 3 ex. Concerned to IoT.

→ The Internet of Things (IoT) refers to idea of connecting everyday physical objects — not general-purpose computers like PCs, tablets or phones, but ordinary things like umbrellas, alarm clocks, medicine bottle caps & bus stops — to internet. Each "thing" is embedded with sensors &/or actuators to collect data from real world & to produce an effect back to real world respectively and uses its internet connection to send data for processing & to receive instructions, all without needing to be a computer itself.

Three examples of IoT:

i) My Train (Alarm Clock): This clock check train timings online overnight; allowing you to sleep if train is delayed instead of waking you up at the normal time. It senses external data (train status) & acts on it (adjusting the alarm).

ii) Glow Cap (Medicine bottle cap): The cap blinks to remind a patient to take their tablets at the right time. If patient forgets, cap connects to internet & emails their doctor to alert them — replacing a single physical reminder with a

networked escalation.

iii) Smart Umbrella: Its handle glows if it has checked an online weather report & predict rain. This is an ambient signal you glance at on your way out.

Q 2- Why is IoT feasible & widespread in today's world?

→ IoT has become feasible & widespread because of converging factors as below:

1. Computing power became cheap: In 80s, a PC with Intel 486 chip costed \$1500. Today, it costs \$1. Cost decrease & performance increase is explained by Moore's Law — the observation that no. of transistors that can fit on a silicon chip roughly doubles every 18 months.

2. Internet Connectivity is now cheap & ubiquitous: When early internet access relied on slow, expensive dial-up connections, most homes today have always-on broadband & Wifi now removes need to run network cables to every device.

3. Online platforms & dev tools have matured: Rise of "Web 2.0" popularised API use, which let programs talk directly to online services. Combined with easy web frameworks & scalable cloud hosting, building "online half" of a connected product is now easy.

Q. 3. What is User Design & Interaction Design?

Explain diff betⁿ two.

→ Design, as per connected devices, is a wider field than just shape/look of an object. Several related design disciplines apply:

- Industrial/Product Design covers form & decoration of an item, and functional aspects such as how product is constructed & how its controls are made easy to understand.
- User (Experience) Design focuses on object's UI - whether that's a screen/physical button & takes end users' perspective as central concern. It creates best possible solⁿ for user, as enjoyable & efficient possible.
- Service Design takes broadest view & looks at the service in its entirety - not just a device - but the whole system of touchpoints a user experiences.
- Interaction Design looks specifically at how diff. parts of a sys. interrelate with each other & especially how user features within that interaction.

Difference: UX design centres individual end user experience of single interface/device - making interaction as great as possible. Interaction design takes a wider-angle view of how the various parts of a whole system relate to one another & how users move through & across that sys.

Q.4. What is Manufactured Normalcy Field? Explain concept.
→ Term Origin: The concept was formed by technology blogger Venkatesh Rao to explain how new tech gets adopted by society.

Concept: He observed that we don't experience present as something that is constantly changing. Big advances "sneak up on us" gradually, hidden in plain sight. The normalcy despite constant change is the concept.

Why it matters for IoT Design: To get a new tech. adopted it must be given in a way that fits inside people's current manufactured normalcy field. A UX designer presents an experience that doesn't stretch boundaries of user's normalcy field, even if underlying tech is advanced, if successful.

Relevance to IoT: IoT bestows hidden, "magical" capabilities on everyday objects. Designers use familiar metaphors to help feel such capabilities a natural small step rather than alien leap = easing adoption.

Example: Mobile was introduced as "a phone you can carry around". Only later it evolved into a portable internet terminal that can play movies, play music & call at same time.

Q.5. Explain TCP & UDP.

- (i) Concept explanation with example.
(ii) Diff betⁿ both

pta →

TCP (Transmission Control Protocol) is a connection-oriented & reliable transport protocol built on top of IP. It guarantees delivery through sequence nos., acknowledgements & retransmission of lost data, so msg. arrive complete & in order - at cost of some extra overhead & delay.
Ex.: Sending an Email or Loading a web page (HTTP)

UDP (User Datagram Protocol): UDP offers "best-effort" delivery. Each msg. may or may not arrive. There is no handshake, acknowledgement, retransmission & no delay caused by waiting for msg. to arrive in sequence. Lack of overhead makes UDP faster & lighter than TCP. Ex:- Voice over IP (VoIP)

Aspect	TCP	UDP
Connection type	Connection-oriented (reliable)	Connectionless (best-effort)
Delivery guarantee	Guaranteed - lost data is retransmitted.	Not guaranteed - a msg. may simply not arrive
Ordering	Msg. are sequenced & reassembled in correct order.	No ordering guaranteed
Acknowledgement	Yes - sender is told what was received	No acknowledgement mechanism
Overhead / Speed	Higher overhead, some delay	Very low overhead, faster
Message length	Can be arbitrarily long	Limited to what fits in a datagram
Typical uses	Email, HTTP	DNS, DHCP